10. DHCP CONFIGURATION

Course: Diploma in Cyber Security

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Introduction

DHCP automatically gives network settings to clients. The common DHCP process is Discover, Offer, Request and Acknowledge. I can practice this in Cisco Packet Tracer without changing a real network.

Task 1: DHCP server and two PCs

Create one DHCP server and two PCs in Packet Tracer. Configure the DHCP pool and set both PCs to obtain addresses automatically. Verify that the PCs receive different IP addresses.

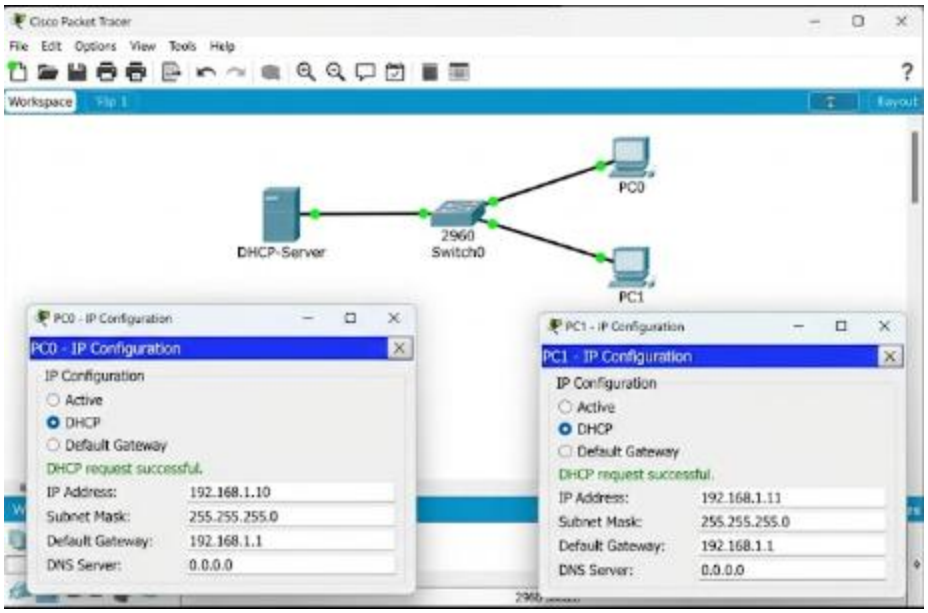


Figure 1: Cisco Packet Tracer Topology and Dynamic IPv4 Allocation on PC0 and PC1

Task 2: DHCP sequence diagram

Draw Client → Discover → Server, Server → Offer → Client, Client → Request → Server, and Server → Acknowledge → Client.

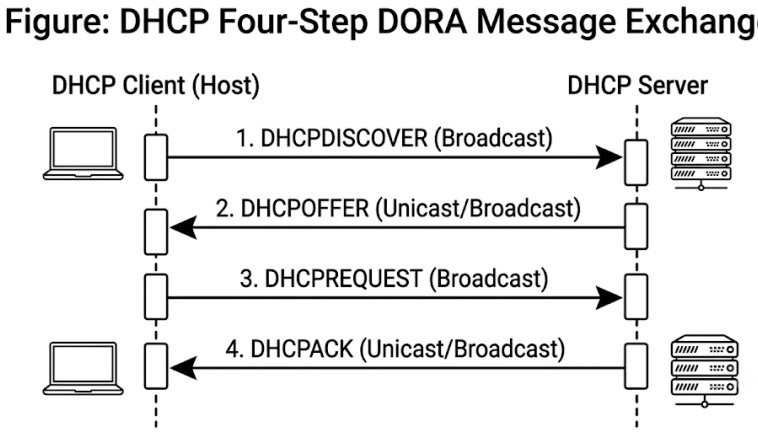


Figure 2: DHCP Four-Step Handshake Protocol Exchange Flowchart (DORA)

Task 3: One-address pool

Configure a DHCP pool with only one usable address and connect two PCs. One should obtain the available address while the other cannot obtain a normal DHCP lease.

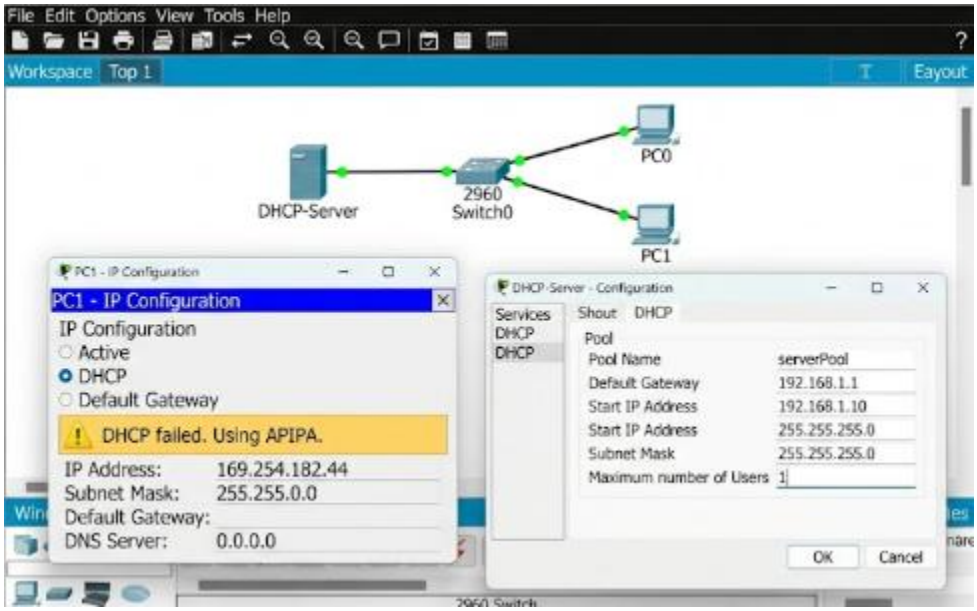


Figure 3: DHCP Pool Exhaustion Resulting in Automatic Private IP Addressing (APIPA) Fallback

Task 4: Restaurant Wi-Fi scenario

Use a laptop and a generic PC representing a smartphone. Configure DHCP so both receive addresses automatically. DHCP is useful because new devices can join without manual IP configuration.

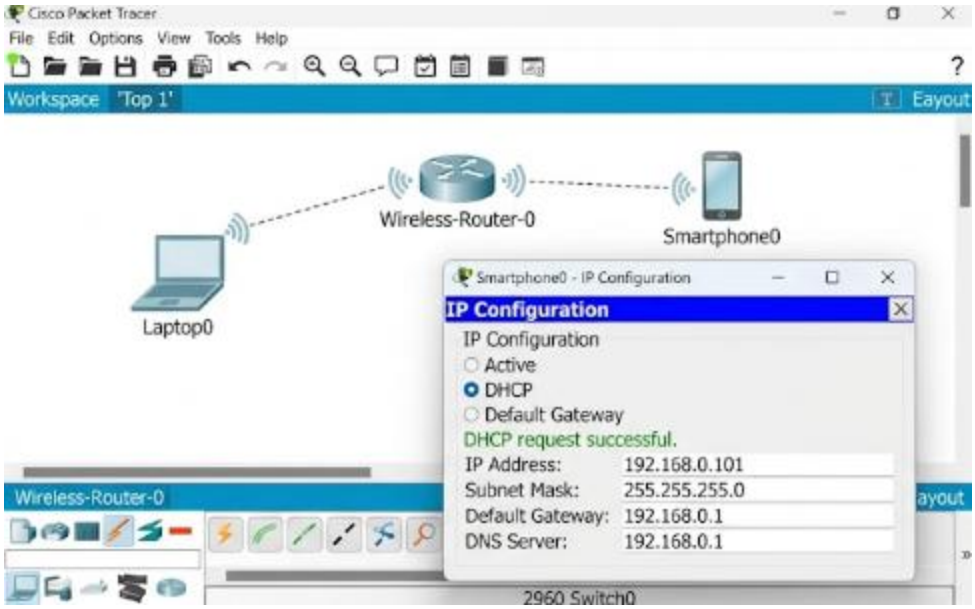


Figure 4: Automated DHCP Provisioning for Guest Devices in a Restaurant Topology

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.